

=====

COMPLAINT2RESOLUTION

MASTER PRODUCT + BUILD PRD

SIH 2026 — SMART AUTOMATION

=====

DOCUMENT PURPOSE

This document is the complete source of truth for building Complaint2Resolution.

The application is a web-based AI-powered civic complaint, accountability and verified-resolution platform.

The goal is NOT to build a basic complaint registration website.

The goal is to build a system that takes a civic complaint from:

SUBMISSION



AI UNDERSTANDING



DEPARTMENT ROUTING



OFFICER ASSIGNMENT



SLA TRACKING



WORK EXECUTION



EVIDENCE SUBMISSION



AI VERIFICATION



CITIZEN VERIFICATION



VERIFIED RESOLUTION



ACCOUNTABILITY SCORE

=====

1. PROJECT NAME

=====

Complaint2Resolution

Tagline:

"Don't Just Register Complaints.

Drive Them to Verified Resolution."

=====

2. PROJECT TYPE

=====

Type:

AI-powered Civic Complaint & Accountability Platform

Platform:

Responsive Web Application

Primary SIH Theme:

Smart Automation

Primary Users:

- Citizens
- Officers
- Department Administrators
- Central Authority

=====

3. CORE PROBLEM

=====

Traditional civic complaint systems generally solve only:

"Can a citizen submit a complaint?"

Complaint2Resolution focuses on:

"Can the system ensure that the complaint reaches the
correct department, gets action within the expected time,
provides evidence of work, gets verified, and remains

accountable until genuine resolution?"

Common problems:

- Citizen does not know which department to contact.
- Complaints may be routed incorrectly.
- Complaint descriptions may be incomplete.
- No mandatory evidence.
- Officers may not update status.
- Citizens may not know what is happening.
- Complaints can remain pending for long periods.
- There may be no automatic SLA monitoring.
- Delayed complaints may not automatically escalate.
- Officers may mark work as completed without sufficient evidence.
- Citizens may submit duplicate complaints.
- Authorities may not have department-level performance visibility.
- Recurring civic issues are difficult to identify.

=====

4. PRODUCT VISION

=====

Complaint2Resolution should make the citizen experience extremely simple while making the government/department workflow intelligent.

Citizen should only need to provide:

1. PHOTO

2. LOCATION

3. DESCRIPTION

The citizen does NOT need to decide:

Which department?

Which officer?

Which category?

What priority?

What SLA?

The system handles those through AI + configured rules.

=====

5. CORE PRODUCT PRINCIPLE

=====

The product is based on:

EVIDENCE

+

AI

+

AUTOMATION

+

SLA

+

ACCOUNTABILITY

+

VERIFICATION

=====

6. TECHNOLOGY STACK

=====

FRONTEND

Next.js

React

TypeScript

Tailwind CSS

shadcn/ui

Lucide React

Framer Motion where useful

BACKEND

Next.js App Router

Next.js Route Handlers / API Routes

Server Actions where appropriate

DATABASE / BACKEND PLATFORM

Supabase

Use Supabase for:

- PostgreSQL database
- Authentication
- Storage
- Row Level Security
- Backend data access

AI

--

Google Gemini API

Gemini is the PRIMARY AI provider.

Gemini will be used for:

- Complaint understanding
- Image + text analysis
- Complaint classification
- Priority recommendation
- Department recommendation
- Complaint summary
- Recommended action
- AI Action Brief
- Resolution report generation
- Before/after evidence analysis

- Citizen dispute evidence analysis
- Duplicate complaint detection
- Recurring issue analysis

IMPORTANT:

Gemini API key MUST remain server-side.

Never expose the Gemini API key in frontend/client-side code.

PDF

Use server-side PDF generation.

Maps / Location

Use:

- Browser Geolocation API
- Map interface
- Latitude
- Longitude
- Human-readable address

Use a free/appropriate map provider for the prototype.

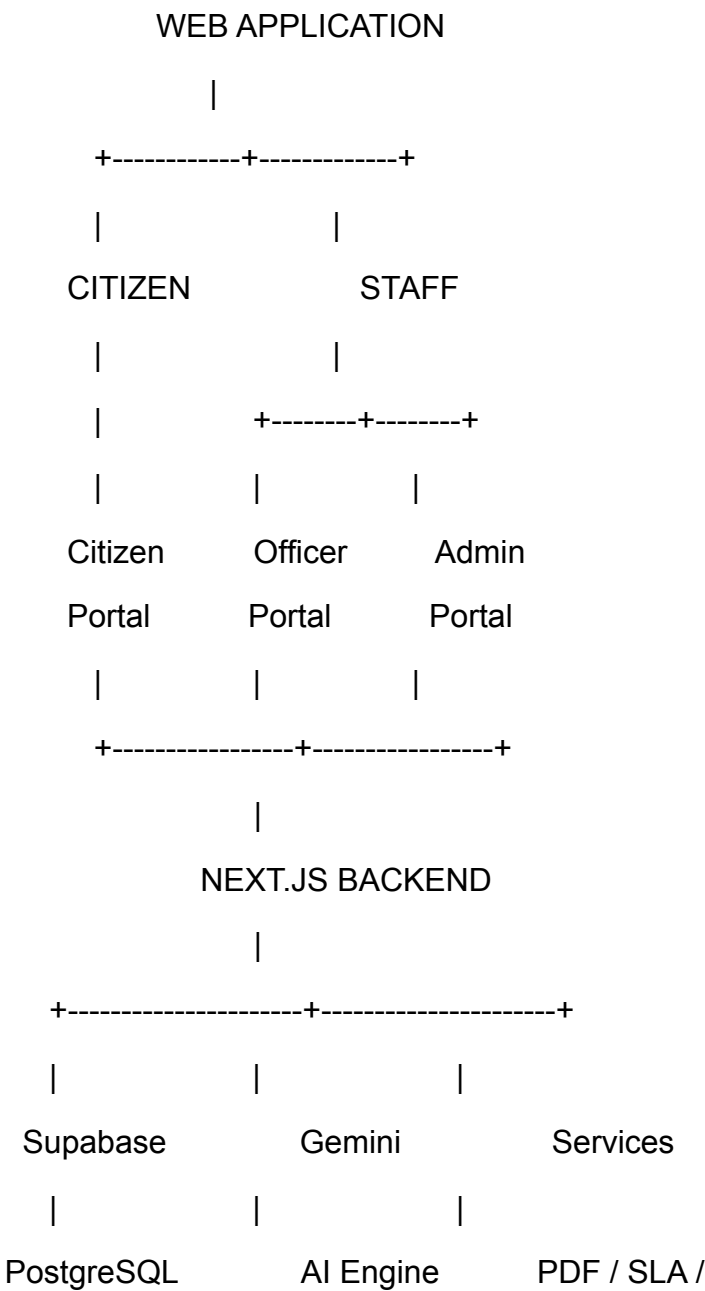
Deployment

Vercel

=====

7. HIGH-LEVEL SYSTEM ARCHITECTURE

=====



Auth	Multimodal	Notification /
Storage	Analysis	Analytics
Row Level Security		

=====

8. USER ROLES

=====

There are FOUR roles.

ROLE 1:

CITIZEN

ROLE 2:

OFFICER

ROLE 3:

DEPARTMENT ADMIN

ROLE 4:

CENTRAL AUTHORITY / SUPER ADMIN

=====

9. AUTHENTICATION RULES

=====

IMPORTANT:

Authentication must remain SIMPLE.

CITIZEN

Citizens can publicly:

SIGN UP

LOGIN

First-time citizen:

Create account



Citizen Dashboard

Returning citizen:

Login



Citizen Dashboard

Citizen signup should be simple.

Recommended fields:

Full Name

Email

Password

Mobile number can be included if required.

Do not make citizen onboarding unnecessarily complicated.

OFFICER

OFFICERS MUST NOT HAVE PUBLIC SIGNUP.

Officer accounts are created internally by:

Department Admin

OR

Central Authority

Officer login:

/officer/login

Officer uses official credentials.

Citizens cannot create officer accounts.

DEPARTMENT ADMIN

Department Admin MUST NOT HAVE PUBLIC SIGNUP.

Account is created internally by:

Central Authority

Login:

/admin/login

CENTRAL AUTHORITY

No public signup.

Configured internally during system setup.

Login:

/admin/login

=====
10. ROLE SEPARATION
=====

Do NOT create one login page where the user chooses:

Citizen

Officer

Admin

Instead use:

Citizen:

/login

Officer:

/officer/login

Admin:

/admin/login

The backend determines the actual role from the authenticated user.

Role permissions MUST be enforced server-side.

Never rely only on frontend role checks.

=====

11. ORGANIZATIONAL STRUCTURE

=====

Do NOT create separate websites/codebases for every department.

Use ONE platform.

Inside the platform create department workspaces.

Example:

CENTRAL AUTHORITY

- |
- +-- Water Management
- |
- +-- Roads / Public Works
- |
- +-- Sanitation
- |
- +-- Electrical
- |
- +-- Drainage
- |
- +-- Parks
- |
- +-- Waste Management
- |
- +-- Other Departments

Each department can have:

- Department Admin
- Officers

Complaint Queue
Categories
SLA Rules
Performance Score

=====

12. CITIZEN LANDING PAGE

=====

The citizen-facing website should be simple.

Main message:

"Report. Track. Resolve."

Primary button:

[REPORT A CIVIC ISSUE]

Secondary button:

[TRACK MY COMPLAINT]

Authentication:

[LOGIN]

[CREATE ACCOUNT]

The landing page should explain:

Report a civic issue



AI understands it



Correct department



Track progress



Verify resolution

=====

13. CITIZEN COMPLAINT SUBMISSION

=====

The complaint form **MUST** have exactly three core mandatory inputs.

MANDATORY FIELD 1:

PHOTO

MANDATORY FIELD 2:

LOCATION

MANDATORY FIELD 3:

DESCRIPTION

The complaint CANNOT be submitted without all three.

=====

14. PHOTO REQUIREMENT

=====

At least one photo is mandatory.

The system should allow:

Camera capture

OR

Gallery upload

Validate:

- File exists
- Valid image format
- Reasonable file size

Optional:

Additional photos

=====

15. LOCATION REQUIREMENT

=====

Location is mandatory.

Citizen can:

Allow GPS

OR

Select location manually on map

Store:

- Latitude
- Longitude
- Address

If location cannot be determined:

Citizen must manually select the location.

No location:

NO SUBMISSION

=====

16. DESCRIPTION REQUIREMENT

=====

Description is mandatory.

Citizen describes:

- What happened
- Where/what is affected
- Any relevant information

Example:

"There has been continuous water leakage outside
the college for four days."

Minimum description length can be configured.

No description:

NO SUBMISSION

=====

17. COMPLAINT VALIDATION

=====

Before submission:

CHECK:

Photo exists?

Location exists?

Description exists?

If NO:

Prevent submission.

Examples:

"Photo is required."

"Location is required."

"Please describe the issue."

=====

18. COMPLAINT CREATION

=====

After successful validation:

1. Upload photo to Supabase Storage.
2. Save complaint.
3. Generate complaint ID.
4. Trigger Gemini analysis.
5. Save AI result.

- 6. Determine department.
- 7. Determine priority.
- 8. Determine category.
- 9. Generate AI summary.
- 10. Generate recommended action.
- 11. Determine SLA.
- 12. Generate complaint PDF.
- 13. Route complaint.
- 14. Set status.
- 15. Notify citizen.

=====

19. COMPLAINT ID

=====

Format:

CR-YYYY-XXXXXX

Example:

CR-2026-001247

The complaint ID is permanent.

If reopened:

SAME COMPLAINT ID

Never create a new complaint ID for a reopened complaint.

=====

20. GEMINI — COMPLAINT ANALYSIS

=====

Gemini receives:

- Citizen description
- Citizen photo
- Location
- Department configuration
- Category configuration

Gemini returns structured JSON.

Expected output:

- category
- subcategory
- department
- priority
- summary
- recommended_action
- suggested_sla_hours
- confidence

=====

21. GEMINI MULTIMODAL ANALYSIS

=====

Gemini must be capable of analyzing:

IMAGE + TEXT

Example:

Photo:

Visible water leakage

Description:

"Water leaking near college for four days."

Gemini should understand both.

The AI should NOT depend only on the description.

The image is evidence.

=====

22. AI OUTPUT EXAMPLE

=====

Example JSON concept:

```
{
  "category": "Water Leakage",
  "subcategory": "Pipeline Leakage",
  "department": "Water Management",
  "priority": "HIGH",
  "summary": "Continuous water leakage reported near XYZ College.",
  "recommended_action": [
    "Inspect the reported pipeline",
    "Identify leakage source",
    "Repair damaged section",
    "Verify leakage has stopped"
  ],
  "suggested_sla_hours": 48,
  "confidence": 0.91
}
```

All AI output must be schema validated before use.

=====

23. AI CATEGORY SYSTEM

=====

Initial categories:

WATER MANAGEMENT

- Water Leakage

- Pipeline Damage
- Water Supply
- Water Contamination

ROADS / PUBLIC WORKS

- Pothole
- Road Damage
- Footpath Damage
- Public Infrastructure

SANITATION

- Garbage Accumulation
- Waste Collection
- Public Cleanliness

ELECTRICAL

- Broken Streetlight
- Exposed Wire
- Electrical Infrastructure

DRAINAGE

- Blocked Drain
- Drain Overflow
- Drain Damage

PARKS

- Damaged Equipment
- Park Maintenance
- Tree/Vegetation Issue

Categories must be configurable.

=====

24. AI DEPARTMENT ROUTING

=====

AI recommends the department.

However:

DO NOT blindly trust AI.

Use:

AI recommendation

+

Configured department rules

+

Location/zone

If AI confidence is high:

Auto route

If confidence is low:

Human review queue

=====

25. AI PRIORITY

=====

Priority:

CRITICAL

HIGH

MEDIUM

LOW

AI should consider:

- Safety
- Severity
- Duration
- Public impact
- Environmental impact
- Infrastructure importance

=====

26. COMPLAINT SUMMARY

=====

AI generates a short professional summary.

Example:

Citizen:

"Water has been leaking outside college for four days."

AI:

"Continuous water leakage reported near XYZ College,
Ward 12, potentially causing water wastage and
public inconvenience."

=====

27. AI RECOMMENDED ACTION

=====

AI generates a practical generic recommended action.

Example:

- 1. Inspect pipeline.
- 2. Identify leakage source.
- 3. Repair damaged section.

4. Verify leakage has stopped.

IMPORTANT:

This is a recommendation.

It is NOT a technical or legally binding instruction.

=====

28. OFFICIAL COMPLAINT PDF

=====

After processing, generate PDF.

PDF must contain:

Complaint ID

Original citizen photo

Location

Coordinates

Citizen description

AI summary

Category

Priority

Assigned department

Recommended action

Submission timestamp

SLA

Resolution deadline

Clearly label:

CITIZEN PROVIDED INFORMATION

and:

AI GENERATED INFORMATION

=====

29. OFFICER PORTAL

=====

Officer login:

/officer/login

After login:

Officer Dashboard

=====

30. OFFICER DASHBOARD

=====

Dashboard sections:

NEW
RECEIVED
ASSIGNED
IN PROGRESS
NEAR SLA
SLA BREACHED
RESOLUTION SUBMITTED
REOPENED

Each complaint card shows:

Complaint ID
Category
Priority
Location
Current Status
SLA Countdown
Deadline

=====

31. OFFICER COMPLAINT VIEW

=====

The officer should see the complaint in a structured workspace.

LEFT:

Complaint PDF
Original Photo
Location
Citizen Description

RIGHT:

AI Action Brief
Priority
Department
SLA Timer
Current Status
Required Actions

BOTTOM:

Timeline
Evidence
Resolution Section

=====

32. AI ACTION BRIEF

=====

The Action Brief is one of the main AI features.

Example:

AI ACTION BRIEF

ISSUE:

Continuous water leakage near XYZ College.

CATEGORY:

Water Leakage

PRIORITY:

HIGH

RECOMMENDED ACTION:

1. Inspect pipeline.
2. Identify leakage source.
3. Repair damaged section.
4. Verify leakage has stopped.

REQUIRED EVIDENCE:

Before Photo

After Photo

SLA:

48 Hours

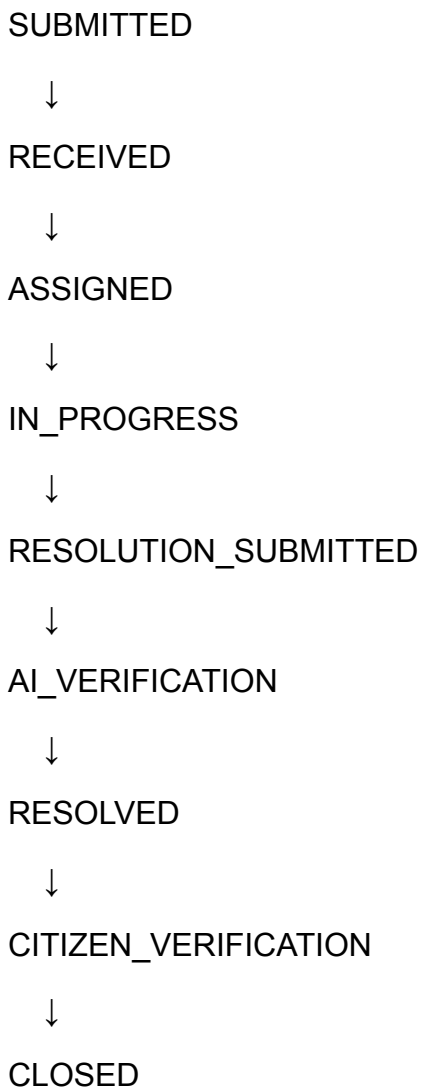
=====

33. STATUS SYSTEM

=====

Use a controlled status state machine.

Main lifecycle:



Dispute flow:

CITIZEN_VERIFICATION



DISPUTED



AI_DISPUTE_VERIFICATION



REOPENED



IN_PROGRESS

Escalation can occur when SLA is breached.

=====

34. AUTOMATIC STATUS UPDATES

=====

When complaint enters department:

Status:

RECEIVED

Citizen sees:

"Your complaint has been received by the department."

When officer accepts/opens/claims responsibility:

Status:

ASSIGNED

Citizen sees:

"Your complaint has been assigned to an officer."

When officer starts work:

Status:

IN_PROGRESS

Citizen sees:

"Work is currently in progress."

When officer submits resolution:

Status:

RESOLUTION_SUBMITTED

=====

35. OFFICER STATUS REQUIREMENT

=====

Officer must update the status as the work progresses.

However, the system must NOT rely only on manual status updates.

SLA monitoring runs independently.

If officer does not act:

Reminder



Warning



Escalation

=====

36. SLA SYSTEM

=====

Each complaint receives an SLA.

Demo defaults:

CRITICAL = 6 hours

HIGH = 24 hours

MEDIUM = 48 hours

LOW = 72 hours

These are configurable.

Production deployment must use official department rules.

=====

37. SLA TIMER

=====

Officer sees:

Deadline:
22 Aug 2026, 10:30 AM

Remaining:
18h 42m

Visual states:

- GREEN
- YELLOW
- ORANGE
- RED

=====

38. SLA REMINDERS

=====

Default:

50% elapsed:

Reminder

75% elapsed:

Warning

90% elapsed:

Critical Warning

100%:

SLA BREACHED

=====

39. SLA ESCALATION

=====

When SLA is breached:

Officer



Department Supervisor



Department Admin



Central Authority

Exact escalation hierarchy must be configurable.

=====

40. ESCALATION RECORD

=====

Every escalation should record:

Complaint ID

Reason

Previous responsible person

Escalated to

Timestamp

SLA duration

Delay duration

=====

41. OFFICER RESOLUTION

=====

Officer cannot simply press:

RESOLVED

without submitting evidence.

Officer must provide:

Action Taken

Before Photo

After Photo

where applicable.

Evidence requirements can be configured by complaint category.

=====

42. BEFORE / AFTER EVIDENCE

=====

Example:

WATER LEAK:

BEFORE:

Leakage visible

AFTER:

Repaired pipeline

STREETLIGHT:

BEFORE:

Broken/non-working light

AFTER:

Repaired light

GARBAGE:

BEFORE:

Garbage accumulation

AFTER:

Cleaned location

ROAD:

BEFORE:

Pothole

AFTER:

Repaired road

=====

43. AI RESOLUTION REPORT

=====

Officer provides a short action note.

Example:

"Damaged pipeline joint repaired."

Gemini generates:

"The reported water leakage was inspected and the damaged pipeline joint was repaired. Post-repair evidence has been submitted for verification."

Officer MUST review the report.

Officer can edit.

Only after officer confirmation can it be submitted.

=====

44. AI RESOLUTION VERIFICATION

=====

Gemini analyzes:

Original complaint photo

Original description

Officer before photo

Officer after photo

Officer action description

Gemini returns:

RESOLUTION_CONSISTENT
POTENTIALLY_UNRESOLVED
HUMAN_REVIEW_REQUIRED

plus:

confidence

=====

45. AI VERIFICATION RULE

=====

Never claim:

"100% verified."

Use:

"Evidence appears consistent with resolution."

or:

"Evidence suggests the issue may remain unresolved."

AI is a decision-support system.

=====

46. CITIZEN RESOLUTION VERIFICATION

=====

After officer submits resolution:

Citizen sees:

RESOLVED — AWAITING YOUR CONFIRMATION

Show:

Original issue

Officer action

Resolution report

Before photo

After photo

Resolution date

Buttons:

[YES — ISSUE RESOLVED]

[NO — ISSUE STILL EXISTS]

=====

47. CITIZEN CONFIRMS YES

=====

Status:

CLOSED

Record:

Citizen confirmed resolution.

Then ask optional feedback:

Rating

Comment

=====
48. CITIZEN SELECTS NO
=====

Do NOT immediately create a new complaint.

Show:

"Please upload a current photo showing why the issue
is still unresolved."

CURRENT PHOTO:

MANDATORY

Optional:

Additional description

=====

49. AI DISPUTE ANALYSIS

=====

Gemini receives:

- Original photo
- Original description
- Officer before photo
- Officer after photo
- Officer resolution report
- Citizen current photo
- Citizen dispute description

=====

50. DISPUTE OUTCOME 1

=====

AI determines:

ISSUE APPEARS UNRESOLVED

Then:

REOPEN EXISTING COMPLAINT

Same:

Complaint ID

Example:

CR-2026-001247

Do NOT create:

CR-2026-001248

=====

51. DISPUTE OUTCOME 2

=====

AI determines:

ISSUE APPEARS RESOLVED

Do NOT reopen.

Do NOT create duplicate.

Show:

"The submitted evidence appears consistent with the
previously reported issue having been addressed."

Show:

Resolution Report

Before Photo

After Photo

Action Taken

Resolution Date

=====

52. DISPUTE OUTCOME 3

=====

AI confidence is insufficient.

Result:

HUMAN_REVIEW_REQUIRED

Supervisor sees:

Original evidence

Officer evidence

Citizen evidence

AI explanation

AI confidence

Supervisor can:

Confirm Resolution

OR

Reopen Complaint

=====

53. REOPENED COMPLAINT

=====

When reopened:

SAME COMPLAINT ID

Status:

REOPENED

Then:

IN_PROGRESS

Notify:

Officer

Department Admin

Citizen

=====

54. RESOLUTION REPORT

=====

Generate a final report containing:

Complaint ID

Original issue

Original location

Department

Officer

Submission date

SLA

Resolution deadline

Actual resolution date

Action taken

Before photo

After photo

AI verification

Citizen confirmation

Feedback

=====

55. CITIZEN DASHBOARD

=====

Show:

- Total Complaints
- Active
- Resolved
- Closed
- Reopened

Complaint cards:

- Complaint ID
- Category
- Location
- Priority
- Department
- Status
- Date

=====

56. CITIZEN COMPLAINT DETAIL

=====

Show:

Complaint ID
Original Photo
Location
Description
AI Summary
Department
Priority
Timeline
Current Status
SLA where relevant
Resolution Report
Evidence
Feedback

=====

57. DEPARTMENT ADMIN DASHBOARD

=====

Show:

Total Complaints
Pending
In Progress
Resolved
SLA Breached
Escalated
Reopened
Average Resolution Time

SLA Compliance

Actions:

Assign Officer

Reassign

Review Complaint

Review Escalation

Review Dispute

View Officers

View Performance

=====

58. CENTRAL ADMIN DASHBOARD

=====

Central Authority sees the complete system.

MAIN METRICS:

Total Complaints

Resolved

In Progress

Pending

SLA Breached

Escalated

Reopened

Average Resolution Time

=====

59. DEPARTMENT PERFORMANCE

=====

Example:

DEPARTMENT PERFORMANCE

Water Management	92/100
Sanitation	87/100
Roads	74/100
Electrical	66/100

Metrics:

SLA Compliance

Resolution Rate

Reopen Rate

Escalation Rate

Average Resolution Time

Citizen Rating

=====

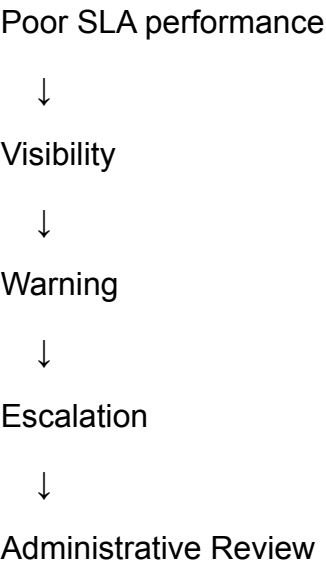
60. ACCOUNTABILITY MODEL

=====

The platform does NOT automatically deduct salaries.

Do not implement payroll functionality.

Instead:



Central Authority can see which departments consistently perform poorly.

The platform provides evidence for administrative action.

Actual disciplinary or financial decisions remain with the authorized authority and applicable rules.

=====

61. OFFICER PERFORMANCE

=====

Metrics:

- Complaints Assigned
- Complaints Resolved
- SLA Compliance
- Average Resolution Time
- Reopened Cases
- Escalations
- Citizen Feedback

=====

62. RECURRING ISSUE DETECTION

=====

Gemini/analytics should detect patterns.

Example:

- 20 streetlight complaints
- Same area
- Same month

System:

"Potential recurring infrastructure issue detected."

Admin sees:

- Location
- Category
- Number of complaints
- Average resolution time
- Reopen rate

=====

63. DUPLICATE DETECTION

=====

Potential duplicates are identified using:

- Location
- Category
- Description
- Image similarity / visual evidence
- Time proximity

Example:

5 citizens report same streetlight.

System:

Potential Complaint Cluster

=====

64. COMPLAINT TIMELINE

=====

Every complaint has a chronological timeline.

Example:

10:30

Complaint Submitted

10:31

AI Analysis Complete

10:31

Department Assigned

10:32

Received

10:45

Officer Assigned

11:10

Work Started

Next Day 14:20

Resolution Submitted

14:21

AI Verification

15:00

Citizen Notified

16:30

Citizen Confirmed

17:00

Closed

=====

65. NOTIFICATIONS

=====

CITIZEN:

- Complaint submitted
- Complaint received
- Complaint assigned
- Work started
- Resolution submitted
- Verification required
- Reopened
- Closed

OFFICER:

- New complaint
- Assignment
- SLA reminder
- SLA warning
- SLA breach
- Citizen dispute
- Reopened complaint

ADMIN:

- SLA breach
- Escalation
- Repeated violations
- High-priority complaint
- Recurring issue
- Department score warning

=====

66. PAGE STRUCTURE

=====

PUBLIC:

/

Landing Page

/login

Citizen Login

/signup

Citizen Signup

/report

Complaint Submission

/track

Complaint Tracking

CITIZEN:

/citizen/dashboard

/citizen/complaints/[id]

/citizen/complaints/[id]/verify

/citizen/profile

OFFICER:

/officer/login

/officer/dashboard

/officer/complaints/[id]

/officer/complaints/[id]/resolve

DEPARTMENT ADMIN:

/admin/login

/admin/dashboard

/admin/complaints

/admin/officers

/admin/analytics

CENTRAL AUTHORITY:

/admin/dashboard

/admin/departments

/admin/escalations

/admin/analytics

=====

67. ROLE PERMISSIONS

=====

CITIZEN:

Can:

- Create own account.
- Create complaints.
- View own complaints.
- View own PDFs.
- View own resolution reports.
- Verify own resolutions.
- Submit dispute evidence.
- Give feedback.

Cannot:

- View other citizens' complaints.
- Access officer portal.
- Access admin portal.

OFFICER:

Can:

- View assigned department complaints.
- Update permitted statuses.
- Add action notes.
- Upload evidence.
- Submit resolution.
- View AI Action Brief.

Cannot:

- Create admin accounts.
- Modify department settings unless authorized.
- Access unrelated departments.

DEPARTMENT ADMIN:

Can:

- Manage department complaints.
- Assign/reassign officers.
- View department analytics.
- Review escalations.
- Review disputes.
- Manage department officers.

Cannot:

- Access unrelated department data unless authorized.

CENTRAL ADMIN:

Can:

- View all departments.
- Manage departments.
- Manage department admins.
- View all complaints.
- View escalations.
- Configure SLA rules.
- View analytics.
- View performance.

=====

68. DATABASE CONCEPT

=====

Use Supabase PostgreSQL.

Core tables:

users / profiles

departments

department_categories

officers

complaints

complaint_images

complaint_ai_analysis

complaint_status_history

complaint_assignments

sla_rules

sla_events
escalations
resolution_submissions
resolution_evidence
resolution_reports
ai_verifications
citizen_verifications
feedback
complaint_clusters
notifications
department_scores
audit_logs

=====

69. SUPABASE AUTH

=====

Use Supabase Auth.

Citizen:

Email/password authentication.

Officer/Admin:

Internal accounts.

Role information must be stored securely.

Use Supabase Row Level Security.

Examples:

Citizen:

SELECT only own complaints.

Officer:

SELECT permitted department complaints.

Department Admin:

SELECT department complaints.

Central Admin:

SELECT all permitted data.

=====

70. SUPABASE STORAGE

=====

Storage should contain separate logical areas:

complaint-images

resolution-evidence

generated-pdfs

resolution-reports

Use secure/private buckets where appropriate.

Never expose sensitive files through unrestricted public URLs.

=====

71. BACKEND SERVICES

=====

Create modular service layers.

COMPLAINT SERVICE:

- createComplaint()
- getComplaint()
- updateComplaint()
- transitionStatus()

AI SERVICE:

- analyzeComplaint()
- generateActionBrief()
- generateResolutionReport()
- verifyResolution()
- analyzeDispute()
- detectDuplicate()
- detectRecurringIssue()

SLA SERVICE:

calculateSLA()
checkSLA()
sendReminder()
createEscalation()

PDF SERVICE:

generateComplaintPDF()
generateResolutionReport()

NOTIFICATION SERVICE:

notifyCitizen()
notifyOfficer()
notifyAdmin()

ANALYTICS SERVICE:

calculateDepartmentScore()
calculateOfficerMetrics()
calculateSystemMetrics()

=====

72. GEMINI SERVICE ARCHITECTURE

=====

DO NOT call Gemini directly from UI components.

Use:

Frontend



Next.js API



Gemini Service



Google Gemini API

Gemini API key is stored in server environment variables.

=====

73. GEMINI FAILURE HANDLING

=====

If Gemini fails:

Complaint **MUST** still be saved.

Do not lose the complaint.

Set:

AI_PROCESSING_FAILED

Then retry.

If retry fails:

Send complaint to human routing/review.

AI failure must never destroy or invalidate a legitimate citizen complaint.

=====

74. AI OUTPUT VALIDATION

=====

All Gemini outputs must be schema validated.

If invalid:

Retry

If repeated failure:

Human review

Never use uncontrolled AI text to modify critical database fields.

=====

75. PDF GENERATION

=====

PDF generation happens server-side.

PDF should be generated after:

- Complaint creation
- AI analysis
- Department routing
- SLA assignment

PDF should contain the original evidence and AI-generated analysis.

=====

76. SECURITY

=====

Implement:

- Supabase Auth
- Row Level Security
- Server-side authorization
- API validation
- File validation
- File size limits
- Secure storage
- Environment variables
- Gemini API key protection
- Audit logging

NEVER expose:

- Gemini API key
- Supabase service role key
- Passwords
- Private internal credentials

=====

77. AUDIT LOGGING

=====

Important actions should be recorded.

Example:

Officer changed:
RECEIVED → ASSIGNED

User:
Officer ID

Timestamp:
20 Aug 2026 10:30

Other audit events:

- Complaint creation
- Assignment
- Reassignment
- Status change
- Evidence upload
- Resolution submission
- AI verification
- Escalation
- Reopening
- Admin changes

=====

78. ERROR HANDLING

=====

Frontend:

Friendly error messages.

Backend:

Structured error responses.

Examples:

Photo upload failed.

Location unavailable.

AI processing failed.

Complaint could not be routed.

PDF generation failed.

Do not show technical stack traces to users.

=====

79. RESPONSIVE DESIGN

=====

The application MUST work on:

Mobile

Tablet

Desktop

Citizen portal must be optimized for mobile first.

Officer and Admin dashboards should prioritize desktop while remaining responsive.

=====

80. UI DESIGN

=====

STYLE:

Professional

Modern
Trustworthy
Clean
Government-grade

Avoid:

Excessive gradients
Gaming UI
Excessive animation
Overloaded dashboards
Unnecessary glassmorphism

=====

81. CITIZEN UI

=====

Citizen should feel:

"This is easy."

Main actions:

Report Issue
Track Complaint

Complaint submission should be visually simple.

Use large clear controls:

 Upload Photo

 Select Location

 Describe Issue

=====

82. OFFICER UI

=====

Officer should feel:

"I know exactly what needs to be done."

Prioritize:

- AI Action Brief
- Complaint PDF
- SLA Timer
- Status
- Required Evidence

=====

83. ADMIN UI

=====

Admin should feel:

"I can see exactly where the system is failing."

Prioritize:

- SLA Breaches
- Escalations
- Department Scores
- Complaint Trends
- Recurring Issues

=====

84. DEMO DATA

=====

Seed realistic data.

Departments:

- Water Management
- Roads / Public Works
- Sanitation
- Electrical

Create:

- Officers
- Department Admins
- Complaints
- Resolved complaints
- SLA breaches
- Escalations
- Reopened complaints
- Department scores

=====

85. DEMO ACCOUNTS

=====

Provide development/demo accounts for:

- Citizen
- Officer
- Department Admin
- Super Admin

Do not hard-code production passwords into frontend code.

=====

86. PRIMARY SIH DEMO

=====

MAIN SCENARIO:

WATER LEAKAGE

STEP 1:

Citizen logs in.

STEP 2:

Citizen uploads water leakage photo.

STEP 3:

Citizen provides GPS location.

STEP 4:

Citizen describes issue.

STEP 5:

Submit.

STEP 6:

Gemini analyzes:

Water Leakage

HIGH priority

Water Management

Summary

Recommended action

STEP 7:

Complaint ID generated.

STEP 8:

Official PDF generated.

STEP 9:

Complaint routed to Water Management.

STEP 10:

Officer opens complaint.

Citizen status:

RECEIVED

STEP 11:

Officer accepts.

Citizen status:

ASSIGNED

STEP 12:

Officer starts work.

Citizen status:

IN PROGRESS

STEP 13:

Officer uploads before/after.

STEP 14:

Gemini generates resolution report.

STEP 15:

Gemini verifies evidence.

STEP 16:

Citizen receives resolution report.

STEP 17:

Citizen confirms:

YES

STEP 18:

Complaint:

CLOSED

STEP 19:

Department score updates.

=====
87. SLA BREACH DEMO
=====

Create a demo complaint with an expired SLA.

System:

SLA BREACHED

Then:

ESCALATED

Admin dashboard:

SLA Breach +1

Escalation +1

=====

88. DISPUTED RESOLUTION DEMO

=====

Officer resolves complaint.

Citizen:

NO — ISSUE STILL EXISTS

System:

Current photo required.

Citizen uploads photo.

Gemini:

Evidence appears unresolved.

System:

REOPEN SAME COMPLAINT

This should be a major SIH demo moment.

=====

89. ALREADY RESOLVED DISPUTE DEMO

=====

Citizen disputes resolution.

AI determines:

Evidence appears consistent with resolution.

System:

Do not create duplicate complaint.

Show:

Resolution Report

Before

After

Officer Action

Resolution Date

=====

90. 3-DAY DEVELOPMENT PLAN

=====

DAY 1

FOUNDATION

- Next.js setup
- Supabase setup
- Auth
- Roles
- Database
- Storage
- Citizen signup/login
- Citizen dashboard
- Complaint form
- Photo upload
- Location

- Description
- Complaint creation
- Complaint ID

DAY 2

AI + OFFICER WORKFLOW

- Gemini integration
- Multimodal complaint analysis
- Category
- Priority
- Department routing
- Summary
- Recommended action
- PDF
- Officer portal
- AI Action Brief
- Status system
- SLA timer
- Escalation

DAY 3

RESOLUTION + ADMIN + POLISH

- Before/after evidence
- Gemini resolution report
- Gemini verification
- Citizen verification
- Dispute workflow
- Reopen same complaint
- Admin dashboard
- Department score
- Analytics
- Notifications
- Demo data
- UI polish
- Deployment

=====

91. DO NOT BUILD IN MVP

=====

Do NOT spend 3-day development time on:

- Native Android/iOS application
- WhatsApp integration
- SMS infrastructure
- Real government APIs
- Payroll integration
- Automatic salary deduction
- IoT

- Hardware
- Advanced GIS infrastructure
- Complex video analysis
- Real government identity verification
- Legal decision automation

=====

92. PRODUCT BOUNDARIES

=====

The application must NOT claim:

"AI guarantees resolution."

Use:

"AI analyzes available evidence and provides a
resolution consistency assessment."

The application must NOT claim:

"AI automatically punishes officers."

Use:

"The system provides transparent accountability metrics
and automated escalation."

=====

93. CORE SUCCESS CRITERIA

=====

The prototype is successful if a judge can experience:

1. Citizen signup/login.
2. Citizen submits complaint.
3. Photo is mandatory.
4. Location is mandatory.
5. Description is mandatory.
6. Gemini analyzes complaint.
7. Correct department is selected.
8. Priority is generated.
9. AI summary is generated.
10. AI recommended action is generated.
11. Complaint ID is generated.
12. PDF is generated.
13. Officer receives complaint.
14. Citizen sees RECEIVED.
15. Officer assignment updates citizen.
16. Officer sees AI Action Brief.
17. SLA countdown is active.
18. SLA breach escalates automatically.
19. Officer submits before/after evidence.
20. Gemini generates resolution report.
21. Gemini analyzes evidence.

22. Citizen receives resolution report.
23. Citizen confirms resolution.
24. Complaint closes.
25. Citizen can dispute.
26. Dispute requires current photo.
27. Gemini analyzes dispute.
28. Existing complaint is reopened if unresolved.
29. Duplicate complaint is NOT created.
30. Admin sees department performance.

=====

94. FINAL AI RESPONSIBILITIES

=====

Gemini is responsible for:

1. Complaint understanding
2. Image understanding
3. Classification
4. Priority recommendation
5. Department recommendation
6. Complaint summary
7. Recommended action
8. AI Action Brief
9. Resolution report generation
10. Before/after analysis
11. Dispute evidence analysis
12. Duplicate detection

13. Recurring issue analysis

=====

95. FINAL AUTOMATION RESPONSIBILITIES

=====

Automation is responsible for:

- Complaint routing
- Complaint ID generation
- PDF generation
- SLA calculation
- SLA countdown
- Reminders
- SLA breach detection
- Escalation
- Status notifications
- Resolution workflow
- Citizen verification
- Reopening
- Department score updates

=====

96. FINAL ACCOUNTABILITY LOOP

=====

The platform creates this loop:

Complaint



Department



Officer



SLA



Action



Evidence



AI Verification



Citizen Verification



Department Score



Administrative Visibility

=====

97. FINAL USP

=====

Complaint2Resolution is NOT:

"A website to submit complaints."

Complaint2Resolution is:

"An AI-powered civic accountability engine that takes a complaint from evidence-based submission to department routing, SLA-controlled action, evidence-based resolution, AI verification and citizen-confirmed closure."

=====

98. FINAL PITCH

=====

Existing complaint systems often answer:

"Was the complaint registered?"

Complaint2Resolution answers:

"Was it understood?"

"Did it reach the correct department?"

"Was an officer assigned?"

"Was action taken within the SLA?"

"Was evidence submitted?"

"Does the evidence support the claimed resolution?"

"Did the citizen confirm it?"

"If not, was the SAME complaint reopened?"

"Which departments are performing well?"

=====

99. FINAL ONE-LINE DESCRIPTION

=====

Complaint2Resolution is an AI-powered civic complaint platform that automatically understands, routes, tracks, escalates and verifies citizen complaints until genuine resolution.

=====

100. FINAL INSTRUCTION TO ANTIGRAVITY

=====

BUILD THE APPLICATION ACCORDING TO THIS DOCUMENT.

Treat this PRD as the SOURCE OF TRUTH.

Do not turn this into a generic complaint CRUD application.

The following features are CORE and MUST WORK:

- Photo + Location + Description

- Gemini AI analysis

- Automatic department routing

- AI summary

- AI recommended action

- Complaint PDF

- Officer portal

AI Action Brief
Status workflow
SLA timer
Automatic escalation
Before/after evidence
AI resolution report
AI resolution verification
Citizen verification
Mandatory dispute photo
AI dispute verification
Reopen SAME complaint
Resolution report
Department scoring
Central Admin dashboard

Prioritize a COMPLETE working end-to-end workflow over adding large numbers of secondary features.

If time is limited:

CORE WORKFLOW > EXTRA FEATURES

The final application must be:

Functional
Responsive
Secure
AI-powered
Demo-ready

SIH-ready

Deployable

The most important demonstration should make the judge understand
within a few minutes:

"This system does not just register complaints.

It actively drives them toward verified resolution."

=====

END OF MASTER PRD

=====